

Kubernetes and Terraform POC Roadmap

Goal: understand how Kubernetes works, how Terraform helps provision the infrastructure, and deploy a public ride-sharing POC with a graphical interface.

Day 1

Set the foundation and make the target concrete.

Do:

- Choose the cloud provider: DigitalOcean
- Create or confirm your account
- Create an API token
- Decide your demo domain or subdomain
- Install terraform, kubectl, helm, and doctl
- Make sure Docker image names and registry strategy are decided

Output by end of day:

- Provider selected
- Cloud credentials ready
- Tools installed
- Clear deployment target with one dev cluster and one public URL

Why:

- This removes ambiguity before you write infrastructure

Day 2

Build the Terraform skeleton.

Do:

- Create terraform/modules
- Create terraform/environments/dev
- Add the cluster module
- Add registry module
- Add namespace module
- Add DNS module if you already have a domain
- Wire them together in terraform/environments/dev

Output by end of day:

- Valid Terraform structure
- terraform plan works

Why:

- This is where you learn how Terraform defines the cloud platform

Day 3

Create the cloud infrastructure.

Do:

- Run terraform init, terraform plan, and terraform apply
- Create the Kubernetes cluster
- Configure kube access
- Verify kubectl get nodes and kubectl get ns
- Create the namespace for the app if Terraform does not already do it

Output by end of day:

- Live Kubernetes cluster
- Reachable from your machine
- Namespace ready

Why:

- This is the first real milestone; infrastructure exists

Day 4

Prepare and validate Helm.

Do:

- Go through each chart in helm/
- Run helm dependency build
- Run helm template
- Fix broken values or template issues
- Ensure service hostnames for MySQL, Redis, RabbitMQ, and Kafka are consistent

Output by end of day:

- Renderable charts for all services
- Known dependency hostnames

Why:

- This prevents deployment-day confusion

Day 5

Install infrastructure dependencies inside Kubernetes.

Do:

- Install MySQL
- Install Redis
- Install RabbitMQ
- Install Kafka
- Verify pods and services
- Update microservice values if service names differ from expected hostnames

Output by end of day:

- Infra dependencies running in cluster
- App values aligned with real infra

Why:

- Your microservices cannot run correctly before these exist

Day 6

Deploy backend services.

Do:

- Deploy centralized-config
- Deploy discovery
- Deploy ride-request
- Deploy location-rider
- Deploy location-saver
- Deploy gateway
- After each deploy, check pods and logs and fix config issues immediately

Output by end of day:

- All backend services running
- gateway exposed publicly

Why:

- This is where the app becomes real in Kubernetes

Day 7

Make the project demoable.

Do:

- Add or deploy a simple frontend/UI
- Connect it to the public backend endpoint
- Test the happy path
- Verify you can demonstrate request ride, rider-side activity, and accepted ride

e

- Capture screenshots
- Write a short architecture summary
- Write a short recruiter-facing README section

Output by end of day:

- Public URL
- Visible UI
- Working demo flow
- Material to show recruiters

Why:

- This turns infrastructure work into a portfolio project

What To Prioritize Every Day

- Day 1: tools and provider access
- Day 2: terraform plan
- Day 3: live cluster
- Day 4: valid rendered charts
- Day 5: infra pods running
- Day 6: app backend running
- Day 7: public demo URL and UI

What Not To Get Stuck On

- Perfecting prod architecture
- Managed service migrations
- Advanced observability
- Autoscaling
- Deep Terraform abstractions

For this POC, shipping matters more.

Success Criteria By The End

- I provisioned the cloud Kubernetes infrastructure with Terraform
- I deployed the microservices with Helm
- I exposed the app through a public URL
- I built a graphical demo for the ride-sharing flow
- I can explain the architecture and deployment path